AI: Strategy + Marketing (MGT 853)

Course Wrap (Session 13)

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Can AI teach us about the past?

AI in Society

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- AI can also help us learn a lot about the world
- Not just the present but also the past $\Longrightarrow$ Ancient Herculaneum in present day Italy

ML Failures and Responsibility

- The Hidden Autopilot Data That Reveals Why Teslas Crash – WSJ

Course Overview

1. AI vs ML

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- ML typically relies on human-generated data to succeed

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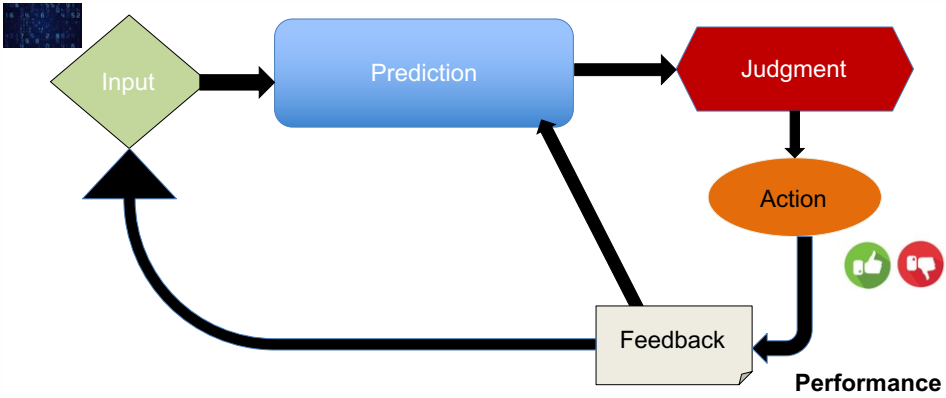
- Generative models like LLMs use self-supervised learning
    - Have prediction at their core (e.g. LLMs with next-token prediction)

AI Decision-Making Framework



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Prediction and Decision

Understand the difference between prediction and decision (not always clear)

Prediction should be an input into your decision, not the other way around.

Which Prediction Problems?

To solve the business problem:

- Are the predictions necessary?
- Are they sufficient?

Consider the Amazon GO Example: Just walk out technology

To solve a business problem as an ML problem

"What prediction when made with 100% accuracy will solve my Business Problem?"

Essential – some (smaller #)

"Nice to have" – Many

$$y = f(x)$$

Start with y

Suitable x?

Y variables

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Which ys are essential?

1. First select the y variables without which business problem *cannot* be solved
2. Typically depends on business problem
3. Some ys can be derived from others, so we don't need to focus on those
4. In Amazon Go, essential ys are 1,7 and 8
   - Others are derivative

X variables (from your submissions)

1. Predicted using x-variables from physical facial attributes (i.e. x1 = distance between eyes, x2 = width of nose, etc.)
2. Items available (volume, SKU, price)
3. Items purchased (volume, SKU, price)
4. Items returned (volume, SKU, price)
5. Customer identity and account information
6. Number of unique customers
7. Amount of time spent by customer in store
8. Day/frequency of visits/purchases
9. Demographics
10. Different items that were picked from the shelves

X variables

1. Person identification
2. Track person movements
3. Detect persons body movement (arms)
4. Detect activity of person
5. Item identification
6. Time spent in the store
7. Distance walked inside the store
8. Average time on each aisle
9. Items grabbed
10. Location point of where they made a stop
11. Past consumption behavior (average $ expenditure per visit, # visits per unit time, etc.)

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- New explainability approaches like GRAD-CAM can help highlight important areas

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- *Confusion matrix* – Do not rely on accuracy as being sufficient. False Negatives NOT the same as False Positives.

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- Use Proxy variables or semi-supervised learning for match happiness

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- Salesforce can lose because of AI (commission on sales of \$30K $\implies$ \$7K)
- Need to *redesign the business model* after AI is introduced

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- Store Managers and AI: Managers can really help AI work better, but have no incentive to help. Need to (re)design incentive structure.

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- Is the method used at the pareto frontier of accuracy / fairness?

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- ML is not magic – has many problems (interpretability)
- Identify and develop complementary sources of (human) value

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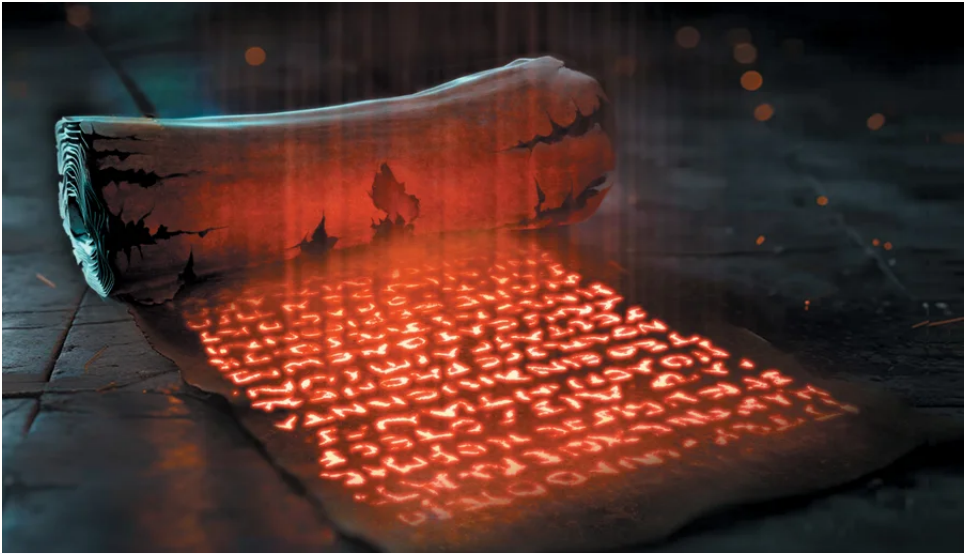

Speaking of the Future...

Course Evaluations

Course Evals

- Now it is time for you to evaluate!
- Please provide detailed feedback at:
- https://students.yale.edu/oce-submissions/